# *16 Designing for fault tolerance*

## This chapter covers

- What fault-tolerance is and why you need it
- Using redundancy to remove single points of failure
- Improving fault tolerance by retrying on failure
- Using idempotent operations to retry on failure
- AWS service guarantees

Failure is inevitable: hard disks, networks, power, and so on all fail from time to time. But failures do not have to affect the users of your system.

A fault-tolerant system provides the highest quality to your users. No matter what happens in your system, the user is never affected and can continue to go about their work, consume entertaining content, buy goods and services, or have conversations with friends. A few years ago, achieving fault tolerance was expensive and complicated, but with AWS, providing fault-tolerant systems is becoming an affordable standard. Nevertheless, building fault-tolerant systems is the top tier of cloud computing and might be challenging at the beginning.

Designing for fault tolerance means building for failure and building systems capable of resolving failure conditions automatically. An important aspect is avoiding single points of failures. You can achieve fault tolerance by introducing redundancy into your system. Instead of running your application on a single EC2 instance, you distribute the